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MILITARY SPECIFICATION

CLEANER, VACUUM, SELF-PROPELLED, 4 X 2 HIGH SPEED, RUNWAY SWEEPING

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high speed, four-wheel, two-wheel drive (4 X 2) self-propelled, runway cleaner, with an air and brush agitation system for vacuum pickup. The cleaner shall be used for cleaning runways, taxiways, and other paved airfield areas required for aircraft operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-S-300	- Sheeting and Tape, Reflective: Nonexposed Lens, Adhesive Backing.
W-B-131	- Battery, Storage, Vehicular, Ignition, Lighting, and Starting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3825

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KKK-T-2108 - Trucks and Truck Tractors: Commercial, Diesel or Gasoline Engine Driven, 24,000 to 32,000 Pounds GVW, 4X2.

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking Blank.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procudures and Tables for Inspection by Attributes.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1223 - Administration Wheeled Vehicles Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The self-propelled runway, vacuum type cleaner, referred to hereinafter as "cleaner", shall be configured with an integral cab and chassis or a commercial cab and chassis with a gross vehicle weight rating (GVWR) minimum of 27,000 pounds (lbs), four-wheel mounted, two-wheel drive (4X2), and shall be diesel-engine-driven. The cleaner shall be furnished complete with cleaning system and control system, and shall be ready for operation.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1, 6.2, and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Safety. The cleaner shall comply with all applicable DoT/OSHA regulations. All parts which are subject to high operating temperatures, or which are energized electrically, shall be insulated, fully enclosed, or guarded when such parts are exposed to contact with personnel or otherwise create a hazard. Exhaust or discharges from the cleaner shall be directed so that they do not endanger personnel. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Protective devices shall not impair the operating functions. Warnings shall be mounted on or near components containing hidden hazards.

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3.6 Ease of maintenance. The cleaner shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the unit. Ease of maintenance provisions shall ensure operating clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls, and threaded fastenings that can be easily over-torqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with substantial, quick-disconnect fastenings. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

3.7 Air pollution control. Vehicle shall comply with EPA regulations in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.8 Performance.

3.8.1 Environmental conditions. The cleaner shall operate satisfactorily in temperatures ranging from minimum of 20 degrees Fahrenheit (°F), to 125°F, when serviced in accordance with manufacturer's recommendations.

3.8.2 Mobility. When fully equipped, the cleaner shall attain the following performance on dry, level concrete or asphalt surfaces:

a. Highway travel at a speed of not less than 50 miles per hour (mph).
b. A turning radius of not more than 24 feet to the left or right as defined in SAE 695b.
c. Ground clearance of not less than 5 inches with pickup nozzle or chamber in the raised position.

3.8.3 Cleaning. The cleaner shall effectively clean runways, taxiways, and similar areas by means of a vacuumized, air and brush agitated pickup system. The cleaning swath shall be not less than 7 feet wide with suction hood only. The cleaner, as a minimum, shall be capable of picking up and retaining the following materials at the rates specified herein.

3.8.3.1 Sand pickup. The cleaner shall pick up commercially procured dry sand, of the 1.5 to 2.5 millimeter (mm) size, from a paved surface covered with a density spread of 0.5 pound per square foot over an area of 140 square feet (7x20 foot spread). Pickup requirements shall be not less than 95 percent of sand (dry weight) at a vehicle speed of 15 mph.

3.8.3.2 Pea gravel pickup. The cleaner shall pick up not less than 95 percent by weight, of pea gravel at a vehicle speed of 15 mph. This requirement shall be represented by a spread of 140 square feet (7x20 feet) with a density of spread of 0.5 pounds per square foot. The graduation of pea gravel size shall be such that 100 percent passes a 3/8 inch screen size and not more than 2 percent passes a U. S. Standard No. 8 sieve.

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3.8.3.3 Stone pickup. At a vehicle speed of 15 mph, the cleaner shall pick up and retain 10 stones having a nominal 2-inch diameter, when placed in two rows of five each at 24 inches apart with the rows 36 inches apart.

3.8.3.4 Solid steel cylinder pickup. At a vehicle speed of 15 mph, the cleaner shall pickup and retain ten solid steel cylinders, 1-inch diameter by three inches long, when placed in two rows of five each at 16 inches apart with the rows 36 inches apart.

3.8.3.5 Joint cleaning. At a vehicle speed of 15 mph, the cleaner shall remove not less than 40 percent by weight of the 1.5 to 2.5 mm dry sand from a rectangular cross section joint 1/2 inch wide by 1/2 inch deep by 6-1/2 feet long when travelling at right angles to the joint.

3.8.3.6 Miscellaneous pickup. At a vehicle speed of 15 mph, the cleaner shall pickup and retain not less than 54 of the items in table I. Seven of the items in table I shall be spaced on a 7 foot by 20 foot level asphalt or concrete pad marked with a 1 foot by 2 foot grid pattern, see 4.5.2.6.

TABLE I. Miscellaneous pickup.

Item	Dimension (nominal)
Steel	
Balls, bearing	1/2-inch diameter
Nails, finishing	2-1/2-inches long
Washers, flat	1/2-inch inside diameter
Capscrews	1/4-inch diameter by 2 inches long
Nuts, hexagon	1/2-inch
Wire, aircraft, safety	1/2-inch long (crumpled)
Aluminum	
Sheet	2 inches square by 1/8 inch thick
Rivets	1/4-inch diameter by 1 inch long

3.9 Chassis. The complete vacuum system, including all components and accessories shall be mounted on a 4 x 2 integral chassis or truck cab and chassis as specified herein.

3.9.1 Integral cab and chassis. Cleaners furnished with integral cab and chassis shall be equipped with but not limited to the following:

a. A diesel engine having the horsepower (hp), torque, and speed characteristics to satisfactorily meet all vehicular performance requirements specified herein on No. 2 diesel fuel. The engine shall be liquid-cooled, compression ignited, two-stroke or four-stroke cycle

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with not less than six cylinders. Engine system shall include an ether starting system or glow plugs. Engine net hp figures used in performance prediction calculations shall be determined in accordance with SAE J816.

b. Full flow-type oil filter shall be furnished.
c. Heavy-duty cooling system is required for the chassis. When specified (see 6.2), silicone rubber hoses shall be furnished with cooling system.
d. Spark arresters shall be provided for engine(s). (Not required on turbo-charged engines.)
e. Suspension, axle, and tire capacities shall be suitable for the intended use.
f. Engine air cleaner shall be dry type.
g. Fuel tank(s) shall have a minimum total capacity of 35 gallons. The tank(s) shall service the chassis engine, and when provided, the fan engine.
h. The cleaner shall be provided with an automatic, heavy-duty, transmission, four speeds forward and one reverse or a hydrostatic drive system with traction motors.
i. Wheels, rims, tires, and tubes (when furnished) shall be in accordance with TRA recommendations. Wheels shall be disk type. Tread shall be highway type. A spare wheel/rim and tire shall be provided (one each size, if front and rear wheels are of different sizes).
j. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations as applicable and as additionally specified herein.
k. The cleaner shall be equipped with power steering.
l. The cleaner shall be furnished with all special tools, if any, that are required for its operation together with the manufacturer's standard tool stowage space or box.
m. A system of rear view mirrors shall be mounted on the right and left outside the cab, which shall give the operator an unobstructed view to the rear on either side of the cleaner. Each mirror shall have a minimum of 50 square inches of flat reflective area and convex surface having a minimum of 20 square inches of reflective area.
n. The cleaner shall be furnished with an audio back-up alarm.
o. The electrical system shall include an electric starter, an alternator, regulator, battery and necessary wiring, fittings, and switches, as specified herein.
p. The cleaner shall be equipped with parking brake capable of being applied and maintained while engine is running.
q. The vehicle's components shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

3.9.2 Truck cab and chassis. Cleaners furnished with truck cab and chassis shall conform to KKK-T-2108, type I, class A and shall be equipped with, but not limited to, the following:

a. The truck shall be powered with a diesel engine having the hp, torque, and speed characteristics to satisfactorily meet all vehicular performance requirements specified herein on No. 2 diesel fuel. The engine shall be liquid-cooled, compression ignited, two-stroke or

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four-stroke with not less than six cylinders. Engine system shall include an ether starting system or glow plugs. Engine net hp figures used in performance prediction calculations shall be determined in accordance with SAE J816.

b. Full flow-type oil filter shall be furnished.
c. Rustproofing is required for the chassis.
d. Heavy-duty cooling system is required for the chassis. When specified (see 6.2), silicone rubber hoses shall be furnished with cooling system.
e. In addition to the exhaust system requirements of KKK-T-2108, spark arresters shall be provided for chassis engine and fan engine. (Not required on turbo-charged engines.)
f. Suspension, axle, and tire capacities shall be not less than 7,000 lb on the front and 20,000 lb on the rear for the chassis.
g. Engine air cleaner shall be dry type.
h. Fuel tank(s) shall have a minimum total capacity of 50 gallons. The tank(s) shall service the chassis engine and the fan engine.
i. Vehicle shall be provided with an automatic, heavy-duty, transmission; four speeds forward and one reverse.
j. Wheels, rims, tires, and tubes (when furnished) shall be the same size for front and rear axles and shall be in accordance with TRA recommendations. Wheels shall be disk type. Tread shall be highway type. No spare tire or wheel is required.
k. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.43 and 393.45 through 393.52, as applicable and as additionally specified herein.
l. The cleaner shall be equipped with power steering.
m. The cleaner shall be furnished with all special tools, if any, that are required for its operation together with the manufacturer's standard tool stowage space or box.
n. A system of rearview mirrors shall be mounted on the right and left outside the cab, which shall give the operator an unobstructed view to the rear on either side of the cleaner. Combination flat and convex mirrors enclosed in a common housing shall be provided. Each combination mirror shall have a minimum of 50 square inches of flat reflective area and convex surface having 20 square inches of reflective area.
o. Cab-to-axle dimension shall be compatible with the cleaning unit, allowing nominal clearances where necessary.
p. The vehicle shall be furnished with a audio back-up alarm.
q. The electrical system shall include an electric starter, an alternator (65 ampere (A) minimum), regulator, battery and necessary wiring, fittings, and switches, as specified herein.
r. Truck shall be equipped with parking brake capable of being applied and maintained while engine is running.
s. The vehicle's components shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

3.9.3 Chassis frame. The chassis frame shall be the manufacturer's standard integral frame or a truck chassis frame. Truck chassis frame shall have heavy-duty main frame or frame shall be reinforced with the reinforcement

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extending at least from the rear of the front suspension rear hanger bracket to the bogie trunnion mounting bracket. The reinforcement shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the loads imposed with the truck loaded to provide the GVWR specified herein. Heavy-duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the GVWR furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. Truck chassis frame rails shall not project beyond the end of the body. Integral chassis frames shall provide sufficient structural strength to resist imposed loads with the hopper filled to capacity.

3.9.4 Service brakes. The cleaner shall be equipped with power-assisted hydraulic or full air brakes on all wheels. Air brake systems, complete with all necessary components, shall include:

a. Air compressor, unloader head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute.
b. Air storage reservoir(s) with not less than 3,000 cubic inches total capacity; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air pressure gage, visible to the driver.
e. Low air-pressure warning, visible and audible.
f. Air control valves.
g. Service brake stop lamp switch.
h. Air dryer shall be provided with automatic moisture ejector and heating device.

3.9.5 Battery(s). The battery(s) provided with the cleaner shall have sufficient capacity to meet the starting and operating requirements specified herein. Each battery shall be a 12 volt (V) battery. The total reserve capacity rating shall be not less than 320 minutes and the total cold cranking rating at 0°F shall be not less than 1200A. Reserve capacity and cold cranking ratings shall be in accordance with SAE J537. The battery(s) shall not be located in the cab. Battery cables shall be protected by use of grommets where the cables pass through metal bulkheads. There shall be sufficient clearances between battery terminals and battery storage containers when connecting cables are in place. The installation of the battery(s) shall provide for easy access for checking the electrolyte without removing the battery(s). When specified (see 6.2), the battery(s) shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

3.9.6 Cab. Manufacturer's standard cab shall be furnished. Tilt cabs shall be furnished with tilting and locking mechanisms. Tilt cab locking mechanism levers shall be readily accessible to the operator while standing on the ground. Cab door(s) shall be equipped with locks, operable from inside the cab. Curbside door, when furnished, shall be equipped with external, key

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operated lock. Cab shall be all steel type with upholstered, adjustable driver's seat or full-width adjustable seat. The door(s) shall open wide, and safety grips shall be furnished on the side of cab, to assist personnel wearing heavy arctic clothing to enter and leave cab. For tilt cabs a grab handle under the windshield and a bumper step, or bumper step cutouts, shall be provided to facilitate cleaning the windshield. Interior lighting shall be provided. Fastenings and seat belts shall be installed. The accelerator pedal shall be physically located perpendicular to the operator's right leg, at an angle of not more than 45°, with a moderate tension level so as not to cause the operator any undue discomfort and/or fatigue.

3.9.6.1 Heater and defroster. Hot water heater shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct heater air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.9.6.2 Windshield wiper(s) and washer(s). The cleaner shall be equipped with windshield wiper(s) and windshield washer(s). Windshield wiper(s) shall be multispeed type and operated by electric motor(s).

3.9.6.3 Air conditioner. When specified (see 6.2), the cleaner shall be equipped with the manufacturer's standard commercial air conditioning unit.

3.10 Cleaning system. The cleaning system shall include a fan, power train, pickup hood and nozzles, air ducts, refuse separator system and a debris hopper. An auxiliary diesel engine may be provided to power the fan and cleaning system. A drain shall be included in the blower compartment to facilitate release of accumulated water and for cleaning/flushing purposes.

3.10.1 Auxiliary engine. When an auxiliary diesel engine is provided, the engine shall develop the necessary brake hp and torque required to drive the fan and all required engine accessories to meet the requirements herein. The engine capacity shall be such that the continuous-duty hp input required by all items driven by the engine will not exceed 90 percent of the full throttle setting of the engine at sea level and at 16° Centigrade (60°F). The engine shall be made for drainage of the fan engine oil pan to a point external of the housing compartment. The engine shall be housed from environmental elements. Engine controls shall be located inside the cab in easy reach of the operator.

3.10.1.1 Safety devices and switches. The auxiliary engine shall be equipped with safety devices for stopping the engine in the event of low lubricating oil pressure or high coolant temperature. The safety devices shall be set as recommended by the manufacturer. The devices shall be adjustable and provided with a visual indicator mounted on the dash of the cab. A starter control safety switch shall be provided to prevent engaging the engine starter while the engine is running.

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3.10.1.2 Air cleaner. A heavy-duty dry type or an oil-bath type air cleaner shall be provided for the auxiliary engine and, if practical, may be mounted on the exterior of the engine housing. The oil-bath type air cleaner shall be so designed that oil is not thrown out of the oil container into or on the engine when the unit is operated over rough terrain. The air cleaner shall have sufficient volume and oil capacity to remove and deposit not less than 2 lbs of dirt and to maintain an efficiency of not less than 95 percent.

3.10.1.3 Governor. An adjustable governor shall be provided and set to limit the engine and vacuum fan speed recommended by the engine manufacturer.

3.10.1.4 Hand throttle. A hand throttle shall be provided for manual control of the engine from idle speed up to and including recommended operating speed. The throttle shall be designed to remain in the setting selected by the operator. The throttle shall be located inside the cab (see 3.12).

3.10.1.5 Hourmeter. An hourmeter shall be provided to record hours of engine operation. The hourmeter shall be mounted on the cab dash panel and shall be located so that it can be conveniently and easily read by the operator.

3.10.1.6 Power train. The power train shall consist of a clutch and all necessary connecting components and bearings for efficient operation. The maximum total angularity of the universal joints shall be not more than 30°.

3.10.1.7 Clutch. Unless otherwise specified (see 6.2), a clutch shall be provided between the auxiliary engine and fan, and shall be compatible with the engine power rating. The clutch control shall be located in the cab and shall be accessible to the driver for engaging and disengaging the fan. When specified (see 6.2), a centrifugal clutch shall be required.

3.10.2 Fan. The fan shall be constructed of lightweight, durable materials mounted on anti-friction bearings, sealed and lubricated for life. The fan or blower shall create air pressure and suction or vacuum for the pickup hood and accessories. The fan, blower housing, and air entering ducts shall be lined or fabricated of wear resistant material.

3.10.3 Pickup hood. A pickup hood assembly shall be provided and shall be situated under the truck chassis (forward or aft of the rear axle), and shall ride on its own set of castor wheels. The pickup hood shall be designed to afford a minimum sweeping width of 84 inches. The hood shall include the necessary piping, hosing, and fittings for connecting to the fan and hopper as necessary and hydraulic lines shall be equipped with quick disconnect fittings to facilitate hood removal. The pickup hood shall be mounted on foam filled or hard, rubber tired wheels or equivalent. The complete pickup hood assembly shall be raised to the travel position or lowered to the work position hydraulically or pneumatically by controls in the truck cab. Ground clearance when the hood is raised shall be not less than 5 inches. A roller brush type

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agitator shall be located within the pickup hood. The design of the pickup hood shall be such that nozzles, skirt/flaps, deflectors, and roller brushes shall not be damaged when passing over a 1-1/2 inch ±1/4 inch obstruction at 15 mph in forward or reverse. If the cleaner cannot be driven in reverse with the pickup hood in the down position, a means shall be provided to automatically raise the hood when the cleaner is placed in reverse gear. An indicator light shall be provided in the cab to alert the driver that the hood is in a lowered position.

3.10.3.1 Agitator roller brush. A hydraulic driven, one piece, agitator roller sweeping brush made of polypropylene bristles and running the full width of the pickup hood shall be furnished. The roller brush shall be at least 12 inches in diameter. A means shall be provided to allow for adjustment of the recommended working pressure or pattern of the roller brush where it meets the pavement being cleaned.

3.10.3.2 Pickup hood seals and deflectors. The pickup hood assembly shall contain rubber seals or brush type rails on the outside edges and include inside deflectors that allow for necessary air flow, channel the dirt and foreign object debris and to create the vacuum space required to pick up the debris. Materials used in the design and fabrication shall be such as to allow for prolonged service life (constant drag with pavement when hood is down) and prevent debris damage to underside components.

3.10.3.3 Side cleaning/sweeping accessories. Unless otherwise specified (see 6.2), laterally directed air flow nozzles shall be fitted on the right and left hand side of the pickup hood to extend the vehicle's cleaning sweeping swath as far as possible to each side and to allow for cleaning around landing lights, pavement depressions, crowned or ramped areas. When specified (see 6.2), air flow nozzles shall be manually adjustable. The air nozzles that are used, the direction of air flow nozzles shall be controlled from inside the truck cab.

3.10.4 Air ducts. Unless otherwise specified (see 6.2), the air ducts shall be constructed of high quality abrasion-resistant steel. When specified (see 6.2), air ducts shall be constructed with steel lined abrasion-resistant rubber compounds. Size and capacity shall be such as to meet the performance requirements specified herein.

3.10.5 Dust and refuse separator. A separator shall be installed in the system to remove dust and refuse from the air stream, before the air passes through the fan, and be deposited in the hopper. Separator shall be designed such that it will not clog or plug with dust and refuse normally encountered, and shall be provided with a cleanout door(s) allowing inspection of its interior. Centrifugal separators, when included in the design, shall be equipped with liners or shall be easily replaceable and shall permit full width inspection. Separator (filtration) system shall be so designed and constructed that it shall operate on either wet or dry pavement without change or modifications during sweeping operations. Filter bags are not acceptable in separator design.

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3.10.6 Dust suppression system. The cleaner shall have a high pressure, low volume water system to suppress dust around the pickup head, and hopper. The system shall be comprised of a high pressure pump or pressurized water system, 200 gallon (minimum) water reservoir, and spray nozzles for fogging action at each location. The system shall be provided with a low or no water warning device. A water hose not less than 25 feet long and 2-1/2 inches in diameter, with storage rack, shall be provided for filling the reservoir. Controls for water valves shall be located in the operator's cab.

3.10.7 Hopper. The hopper, which serves as a collection point for dust and refuse, shall be all steel welded construction with a minimum capacity of 6 cubic yards. The hopper shall be hydraulically dumped through a hydraulically operated or gravity activated gate located so that sweepings will be discharged behind the rear wheels. Design shall insure that the hopper empties completely when in the raised position. Door(s) shall be provided to permit cleanout and inspection of refuse level in the hopper. A support brace or arm shall be provided to hold the hopper door in an open position to facilitate dumping and cleaning of the hopper.

3.10.8 Magnet. A permanent type magnet, having a minimum length of 84 inches shall be mounted in front of the vehicle. The hydraulic, adjustable mount shall afford a magnetic working range above ground level of 2 inches and a traveling position of 5 inches above ground minimum. The holding mechanism for these adjustable mounts shall be rigidly affixed to the truck frame. The adjustable portion of the lift may be either slide or cantilever design but must hold the magnet in constant transverse to travel alignment to the surface on which the truck rests. The magnet shall be capable of demagnification to discharge material it has picked up. The total width of this assembly shall be 96 inches maximum in length and centered along the longitudinal axis of the chassis frame. Controls for the adjustable mount, magnification and demagnification of the assembly shall be located in the operators cab. Indicator lights, red, down and green, up shall be easily seen by the operator. Minimum strength for the magnet shall be 2 inches clearance, 500 gauss magnetic strength; 3 inches clearance, 300 gauss magnetic strength, and 4 inches clearance, 200 gauss magnetic strength.

3.11 Hydraulic system. The hydraulic system shall be compatible with the performance requirements of the cleaner. The system shall consist of a pump driven by the cleaner power unit, a reservoir, suction line strainer, replaceable type full-flow oil filter, necessary hydraulic cylinders, drive motors, control valves, relief valve, hoses, piping, and standard fittings. Hydraulic motors shall drive accessory equipment as specified herein. The cleaner shall be equipped with sufficient controls for hydraulically raising and lowering the nozzle or pickup hood, dumping the hopper, and driving accessory brushes. The controls shall incorporate a float position. The controls shall be so located as to be readily accessible to the operator while seated in the normal position. The control levers shall be so spaced as to permit ease of operation. A minimum number of fittings, joints, and connections shall be used to prevent excessive back pressure, clogging, and leakage. All parts shall be insulated to prevent detrimental vibration and

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shock. The hydraulic pump capacity shall be adequate to provide satisfactory operation of all hydraulically-operated equipment specified herein. Hydraulic tubing shall be of adequate size to permit free flow of the hydraulic fluid at temperatures down -20°F.

3.11.1 Hydraulic system; valves, pistons, and lines. Under maximum pressure conditions, the duty-cycle hydraulic pressures at any point in the system shall not exceed the manufacturer's rated capacity. All hydraulic components shall be designed for easy removal and replacement. Pistons shall be covered or protected to provide for operation throughout the climatic environment. Hydraulic metal tubing shall be used where flexing is not encountered during operation. Armored or otherwise protected rubber or synthetic hose shall be used where flexing is encountered. Provision shall be made for bleeding the system at all necessary points.

3.11.2 Hydraulic supply tank. A hydraulic supply tank shall be provided. The tank shall have the capacity to hold the maximum volume of fluid that will be required in the system during any operating conditions plus a minimum of 10 percent additional air space to handle expansion, foaming, and surging of fluid in the tank. The tank shall be provided with a filler opening of not less than 1-1/2 inches in diameter. The filler opening shall be easily accessible for filling the tank. A vented filler cap shall be provided that will prevent oil from splashing out of the tank during movement. An oil strainer and shutoff valve shall be provided to facilitate cleaning of the fluid and tank. The drain shall be so located that oil will not drain to other components and to allow recovery of the oil in a suitable container. The tank shall be marked "HYDRAULIC FLUID."

3.12 Operating controls. All cleaner operating controls, with the exception of the hopper dump gate control, shall be mounted inside the cab readily available to the operator and shall include:

a. Auxiliary fan engine ignition switch, throttle, tachometer, oil presssure gage, water temperature gage, and hour meter.
b. Pickup hood controls.
c. Electrical system ammeter or discharge lights.
d. Controls for the dust suppression water system, and low/no water warning device.
e. All lighting controls.
f. Side air nozzle accessory controls.

3.13 Lighting system. The cleaner shall be equipped with a lighting system for night operation and highway travel. All lights shall operate from the electrical system of the vehicle. The lighting system shall be in accordance with DoT Federal Motor Carrier Safety Regulations. The system shall include not less than two sealed-beam headlamps with upper and lower driving beams; one adjustable 75,000 minimum candela (candlepower) sealed-beam spotlight located on left side of cab and controllable from driver's side; two combination stoplights and taillights, and emergency flashing lights front and rear (amber). Also two adjustable work lights shall be provided to monitor

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the night use of the units right and left side and two work lights shall be mounted at the rear of the vehicle for use when dumping hopper at night. The lights shall be a minimum of 4 inches in diameter and shall be sealed beam type with white lens. The lights shall be mechanically adjustable from the installed position and shall be controlled (on-off switch) from the cab. Running lights shall be positioned so as not to cast a glare around the windshield area that could distract the operator during night operation.

3.13.1 Warning light. A 12V direct current, rotating warning light with height and diameter measurements not less than 8 inches shall be installed on the top of the vehicle's cab roof. The light shall have an amber lens. A switch shall be provided on the instrument panel to control the light. The light shall be visible 360° around the vehicle.

3.13.2 Turn signals. Turn signals and turn signal unit furnished shall be manufacturer's standard for this type of cleaner.

3.14 Cleaning, treatment, and painting. The exterior surfaces of the cleaner body, truck cab and chassis shall be cleaned and treated in accordance with MIL-STD-1223. The exterior surfaces to be painted shall be primed with a zinc chromate epoxy applied to a dry film thickness of 1.5 to 2.5 mils. The primer used shall be compatible with the final paint system. The finish paint shall be polyurethane enamel, or equal, applied to a dry film thickness of 1.8 to 2.2 mils. The final color shall be yellow gloss No. 13538 of FED-STD-595. The polyurethane enamel shall be equal to DUPONT "IMRON", DITZLER "DURETHANE", OR SHERWIN-WILLIAMS "POLASOL" brand products.

3.15 Identification marking and nameplates. Identification marking and nameplates shall be in accordance with MIL-STD-1223 as required for the Navy. Transportation data plates, conforming to MIL-P-514, type III, composition A or C, shall be furnished and shall be inscribed with all pertinent transportation data, including a silhouette of the cleaner showing the center of gravity.

3.15.1 Safety marking. The cleaner shall be striped front and rear with diagonal safety marking. The stripes shall be alternative 4-inch bands of black and yellow inclined 45° to left and right of the center vertical starting pattern. The black band shall be fabricated from black reflectorized tape conforming to type II, class I, reflectivity 1 of L-S-300.

3.15.2 Instruction plates. The cleaner shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.16 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance

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with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts a suitable warning shall be affixed to the equipment in a conspicuous location.

3.17 Servicing and adjusting. Prior to acceptance of the cleaner by the Government, the contractor shall service and adjust the cleaner for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of chassis and auxiliary engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system(s) filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol). The cleaner shall be conspicuously tagged to identify the lubricants and their temperature range.

3.18 Workmanship.

3.18.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape. All metal fabrication shall be constructed, supported, and braced to prevent any excessive movement caused by equipment vibration, in order to eliminate noise and the abrasive rubbing of hoses and other conduits.

3.18.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.18.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall

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be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.18.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings' ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on cleaner when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one cleaner. All cleaners offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a

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complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each cleaner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. The cleaners shall be tested as specified herein.

4.5.1 First article tests. The first article shall be subjected to the tests specified in 4.5.1.1 through 4.5.2.6. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all cleaners until corrective action has been taken. Acceptance of the first article shall not constitute a waiver by the Government of its rights under provisions of the contract.

4.5.1.1 Run-in test. The cleaner shall be driven at speeds varying from 5 to 15 mph for a total of not less than 10 miles. During this test, the brakes and all hydraulic system components shall be operated. Hopper controls shall be operated with cleaner in a stopped position.

4.5.1.2 Mobility tests. The fully loaded cleaner shall be driven on a paved road. Additionally, the minimum right and left turning radii as well as the ground clearance shall be measured. Failure to meet the requirements of 3.8.2 shall be cause for rejection.

4.5.1.3 Brake tests. In a loaded condition the cleaner shall make not less than five sudden successive stops from a speed of 20 mph. The cleaner shall come to a complete stop within 30 feet. The parking brake shall be tested to demonstrate holding capability.

4.5.2 Cleaning tests. Cleaning tests shall be conducted within an ambient temperature range of 32°F to 100°F. Test areas shall be thoroughly cleaned through the use of hand brooms and hand vacuum cleaners prior to placement of a weighed amount of materials for pickup tests. Test surface area shall not be sealed, polished, or otherwise specially prepared. The requirements of each individual cleaning test specified herein shall be met by one pass of the cleaner over the test area at the stipulated speed. If the cleaner wets the test area, through the use of water during a cleaning test run, then the test area shall be allowed to dry after the sweeping pass. After the area has dried, the material remaining thereon shall be picked up through the use of

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hand brooms and hand vacuum cleaners. The picked up material shall then be weighed so that the percentage of cleaned matter retained by the vehicle can be determined. Cleaning tests shall be conducted with suction head only.

4.5.2.1 Sand pickup test. Commercially procured dry sand shall be spread uniformly over paved, dry, clean surfaces, 7 feet wide and 20 feet long. Using the procedures specified herein, the area shall be cleaned and percentage of pickup determined. Failure of the cleaner to meet the provisions of 3.8.3.1 shall be cause for rejection.

4.5.2.2 Pea gravel pickup test. Pea gravel shall be uniformly spread over a cleaned, paved area measuring 7 feet wide and 20 feet long. Using procedures described herein, the cleaner shall make a 15 mph pass over the test area. Failure to meet the provisions of 3.8.3.2 shall be cause for rejection.

4.5.2.3 Stone pickup. Ten stones, approximately spherical in shape and 2 ±1/4-inch nominal diameter, shall be placed on level concrete pavement in two straight lines 36 inches apart with five stones at 24-inch intervals in each line. The cleaner shall pass over the lines of stones at 15 mph in a direction parallel with the lines. The cleaner shall pick up and retain all of the stones in one pass.

4.5.2.4 Steel cylinder pickup. Ten solid steel cylinders 1 ±0.010 inch in diameter and 3 ±0.010 inch in length shall be arranged as specified on level concrete pavement to perform this test. Two straight lines spaced 36 inches apart shall be formed. Each line shall contain five solid steel cylinders spaced at 16 inch staggered intervals. The solid steel cylinders shall be laid on their sides in random directions. The cleaner shall pass over the lines of cylinders at 15 mph at a direction parallel to the lines. The cleaner shall pick up and retain all the cylinders in one pass.

4.5.2.5 Joint cleaning. A rectangular cross section joint approximately 1/2-inch wide, 1/2-inch deep, and 6-1/2 feet long shall be filled with dry building sand (1.5 to 2.5 mm), level with the surface of the pavement. The cleaner shall meet the pickup requirements of 3.8.3.5 when traveling in a direction perpendicular to the joint.

4.5.2.6 Miscellaneous pickup. Seven each of the items specified in 3.8.3.6 shall be placed on a 7 foot x 20 foot level asphalt or concrete pad that has been marked with a 1 foot x 2 foot grid pattern (1 foot across the 7 foot length and 2 foot down the 20 foot length). The cleaner shall pass over this grid pattern, containing the miscellaneous items, at 15 mph and shall pick up and retain not less than 54 of the items.

4.5.3 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

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4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Unless otherwise specified (see 6.2), the complete cleaner shall be preserved in accordance with the supplier's standard practice. When specified (see 6.2), the complete cleaner shall be preserved and packaged in accordance with the (mobile) requirements of MIL-STD-281, with the level of preservation and packing as specified by the procurement activity.

6. NOTES

6.1 Intended use. Cleaners covered by the specification are intended for use in cleaning runways, taxiways, parking areas, and other airfield paved areas required for aircraft operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
c. When silicone rubber hoses are required (see 3.9.1 and 3.9.2).
d. When maintenance free type battery is required (see 3.9.5).
e. When air conditioner is required (see 3.9.6.3).
f. When manual clutch is required (see 3.10.1.7).
g. When centrifugal clutch is required (see 3.10.1.7)
h. When laterally fixed air flow nozzles are required (see 3.10.3.3).
i. When manually adjustable air flow nozzles are required (see 3.10.3.3).
j. When air ducts are contructed by abrasion-resistant steel is required (see 3.10.4).
k. When air ducts are constructed of abrasion-resistant rubber compounds are required (see 3.10.4).
l. When preservation and packing in accordance with MIL-STD-281 is required and the level of preservation and packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27-410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Blower
Chassis
Engine
Hydraulic systems
Performance

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:
Navy - YD

(Project 3825-N177)

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